Gale UAS: Case Study

UASE 501: Unmanned Aircraft Design

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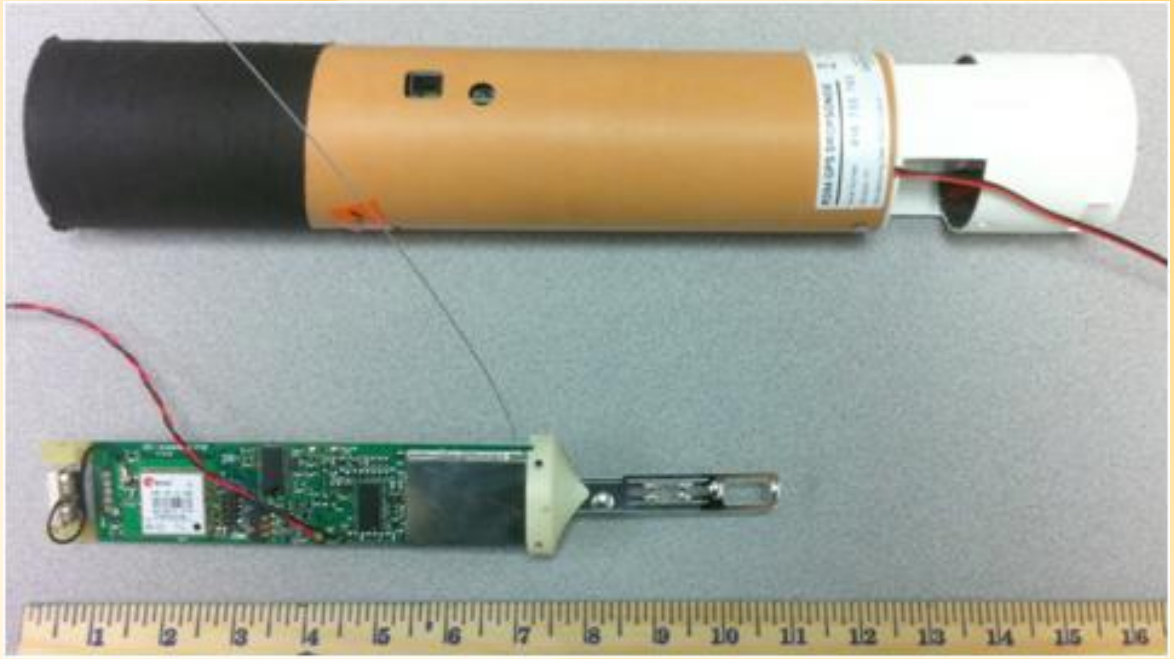
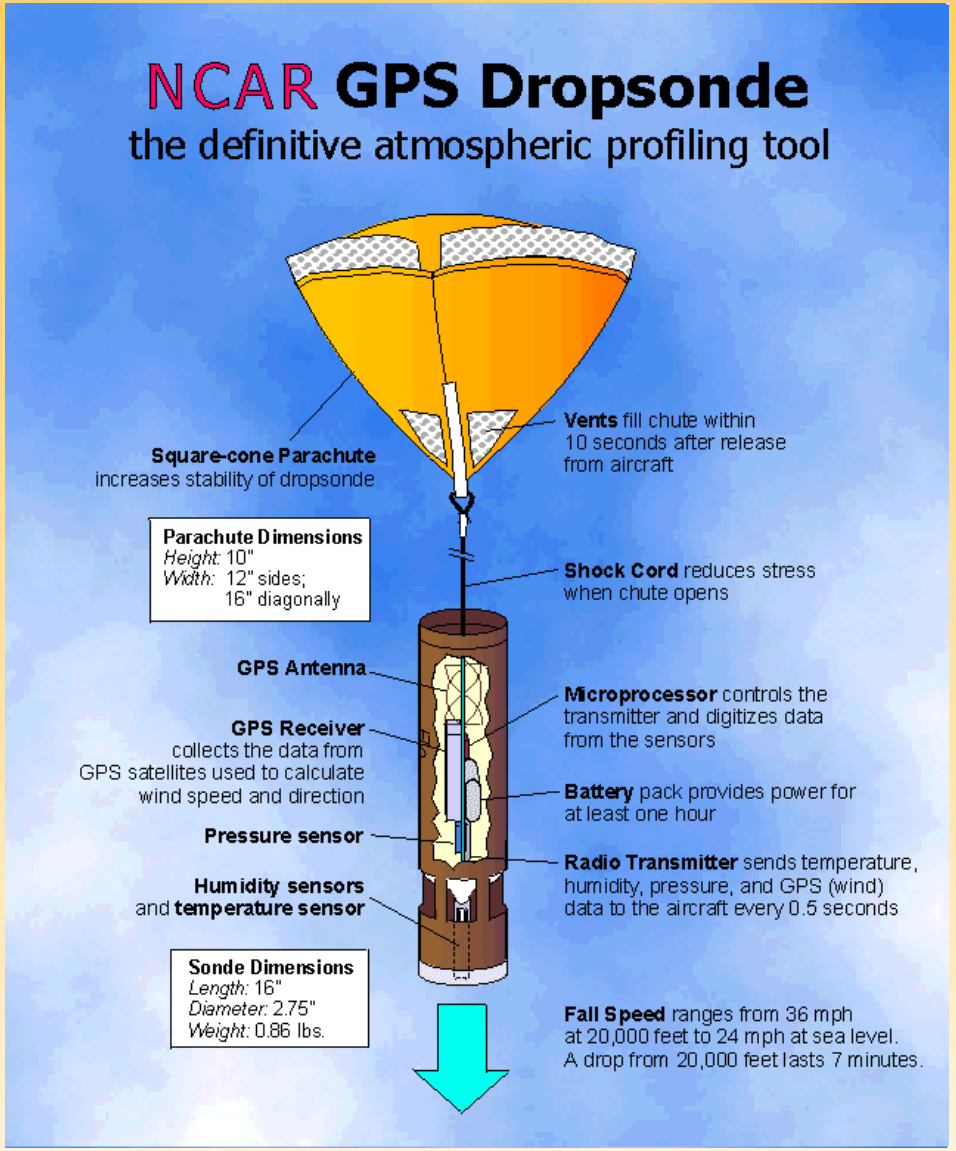
Lecture Outline

- Gale Mission Overview
- Airframe and propulsion
- Avionics and Communication
- Payloads
- Flight Testing
- Post-test system upgrades.

Payload: GPS Dropsond

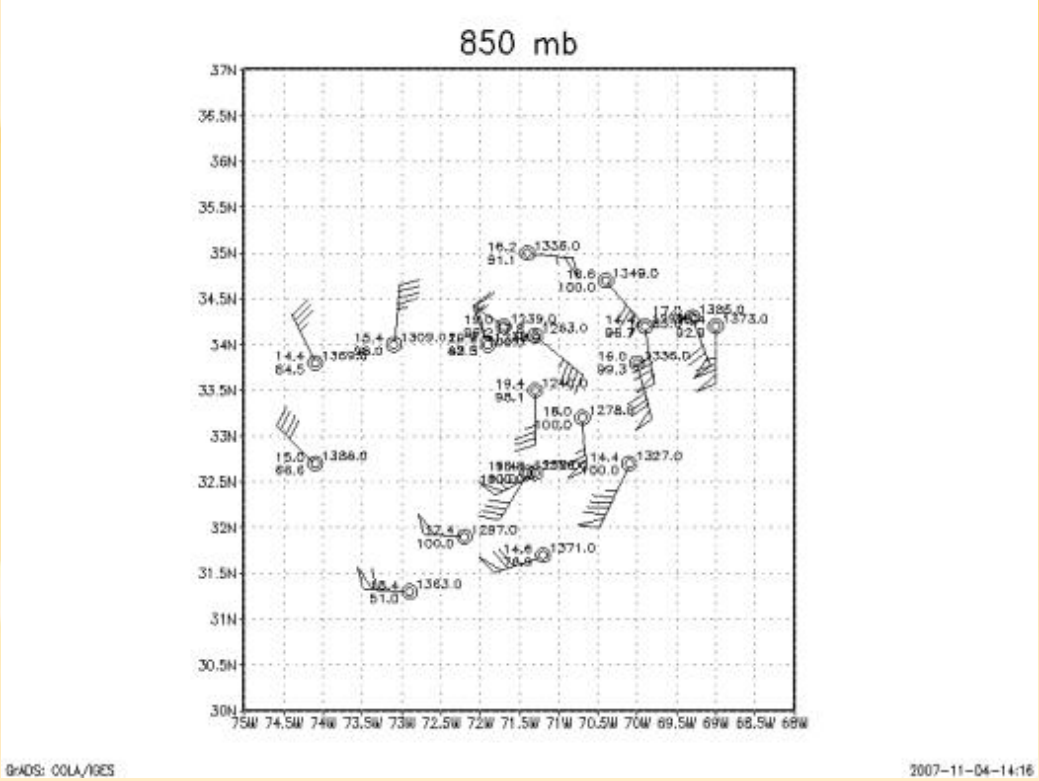




Dropsonde and Hurricane Hunters

Background

- Previous NOAA successes with UAS in hurricanes



Tropical Storm Ophelia, 2005

Hurricane Nole, 2007

Aerosonde Mk II Hurricane Missions

Other Hurricane UAS: Aerosonde

Aspect	Details
(a) System Developer	Aerosonde Pty Ltd (now Textron Systems)
(b) Sponsor	NOAA (in conjunction with NASA for some deployments)
(c) Storm Flown	* Tropical Storm Ophelia (2005, flown before becoming a hurricane) * Tropical Storm Noel (2007)
(d) Date Flown	* Ophelia: 2005; * Noel: November 2, 2007
(e) Summary of Results	• Tropical Storm Ophelia (2005): This was the first deployment of an Aerosonde into a tropical cyclone. It gathered data in the storm's boundary layer, at altitudes as low as 1,200 feet (366m), which is too dangerous for manned aircraft. • Tropical Storm Noel (2007): The Aerosonde successfully flew into the eye of the storm, encountering hurricane-force winds. It reached a low altitude of 300 feet (91 meters). The UAS encountered 80 knots (40 m/s) of wind during its flight. • General Results: The Aerosonde deployments aimed to gather data from the lower parts of hurricanes, documenting and analyzing mesoscale circulations and improving understanding of the high-wind boundary layer. The data included temperature, wind speed, and atmospheric pressure. This data helps in better predicting the development of tropical depressions and storms into hurricanes.

Background

- ERAU began work with NOAA's UAS Hurricane Research Division (HRD) out of Miami in 2009
  - NOAA project sponsor – Joe Cione
  - NOAA outreach and science – Sara Summers
- Goal
  - Provide WP-3D Orion deployable UAS that can be launched from one of NOAA's Hurricane Hunter P-3s into a tropical cyclone
  - Provide in situ measurement of meteorological data
  - Collect boundary layer data at low altitudes
  - Generate a vertical profile of atmospheric data

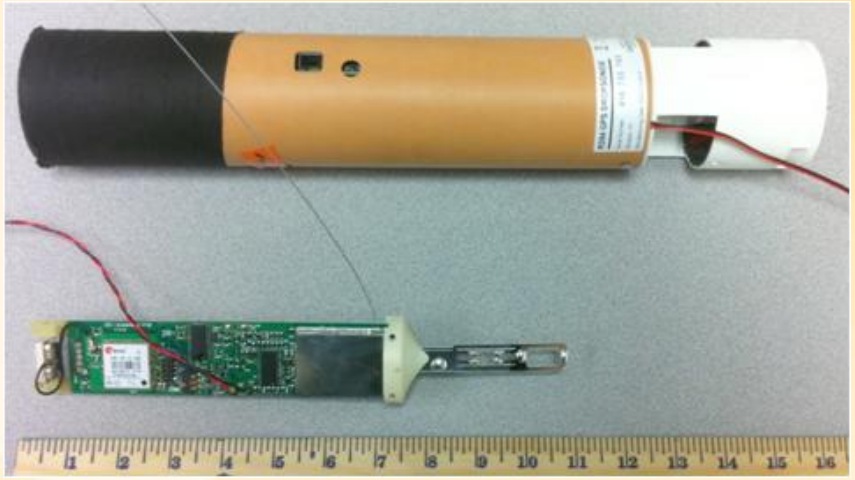
- **Summary: Create a smart flyable dropsonde**

Background

- Challenges to Aerosonde Campaign:
  - Airspace access
    - Previous Aerosonde mission ended with loss of aircraft in order to collect further data in storm
    - Difficult to obtain a Certificate of Authorization
    - Opportunities in:
      - Naval Warning Area off of Key West in Gulf of Mexico
      - Barbados in non-FAA flight information region
  - Long ingress/egress – costal launch and transition to storm presented challenges
    - Greater likelihood of aircraft loss / damage due to high winds
    - Much of vehicle's endurance lost in transition
    - Aerosonde is supposed to be recoverable, which means operation at very low altitudes was discouraged
  - Cost
    - Cost of platform
    - Insurance underwriting
    - Operational costs – flown by a team from AAI out of Australia

Key Requirements

- Shall deploy from P-3 free-fall chute into hurricane and operate for 60 min
- Shall acquire data from the sensor board at a rate of 2Hz for:
  - Air pressure with an accuracy of +/- 0.5 mb or better
  - Air temperature with an accuracy of +- 0.5 C
  - Relative humidity with an accuracy of +/- 5%
  - GPS position with error < 10 meters RMS
  - 3-axis GPS velocity with error of 0.5 m/sec RMS

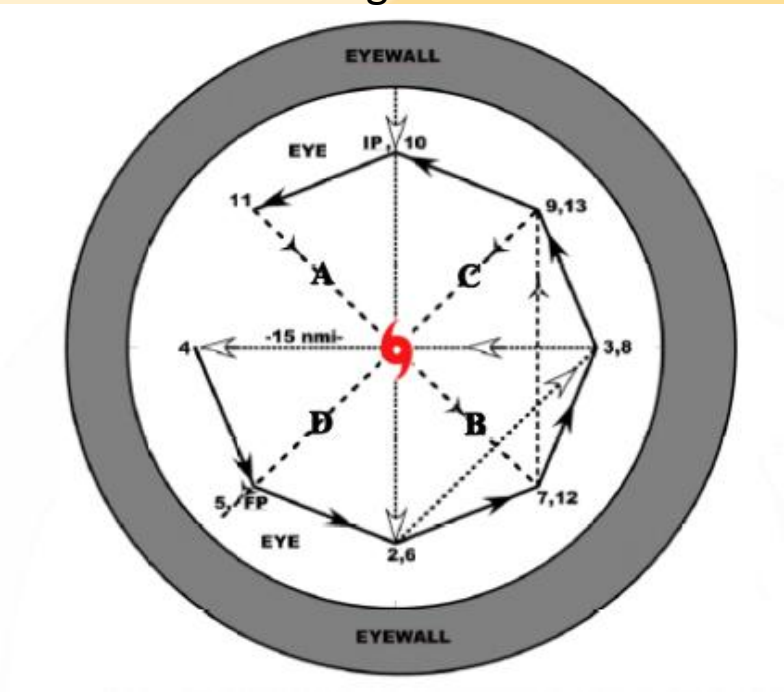


Hurricane Eyewall CONOP #1

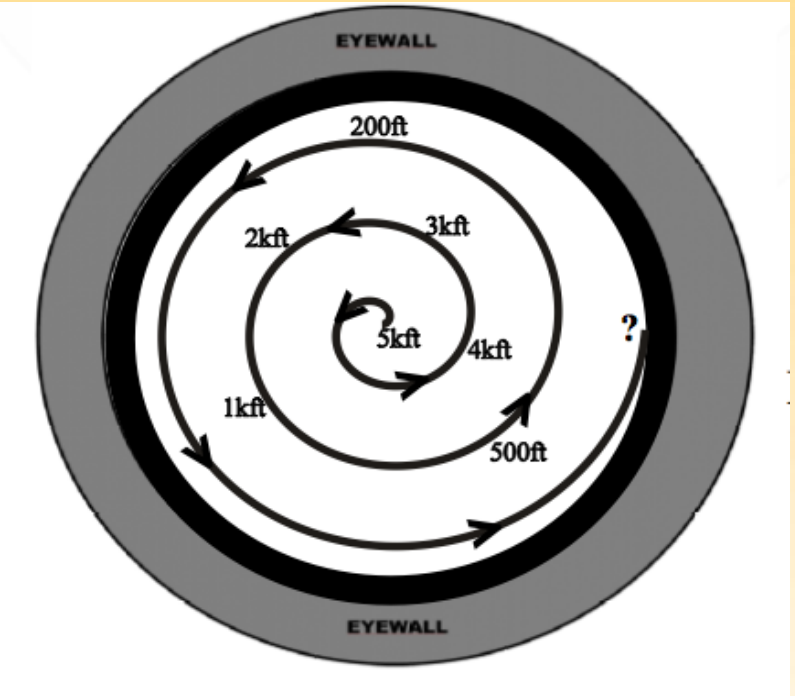
Launch at 10,000 ft. Enter mission flight regime by 5,000 ft.

Spiral, Step Descent through storm until reaching 200 ft. Then, enter eyewall.

P3 – Flight Path



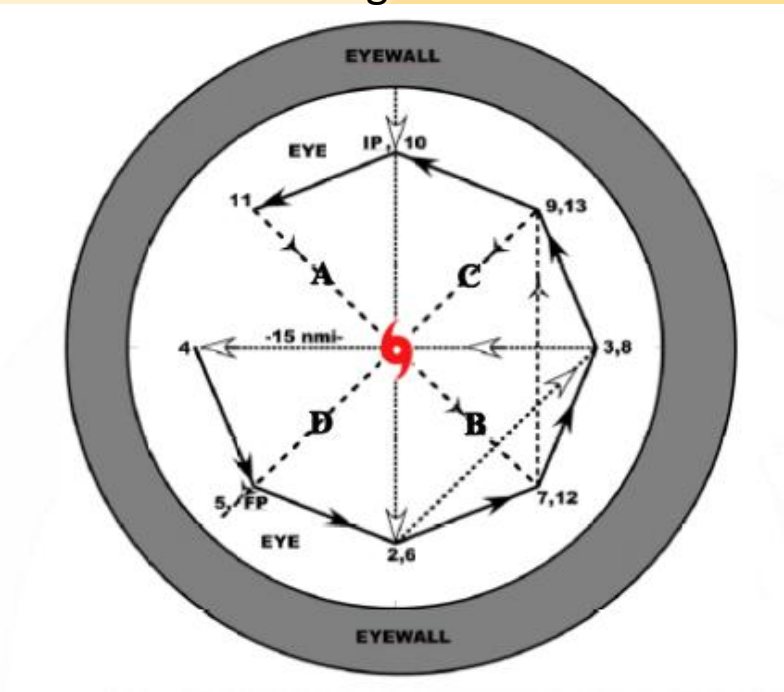
Gale UAS – Flight Path



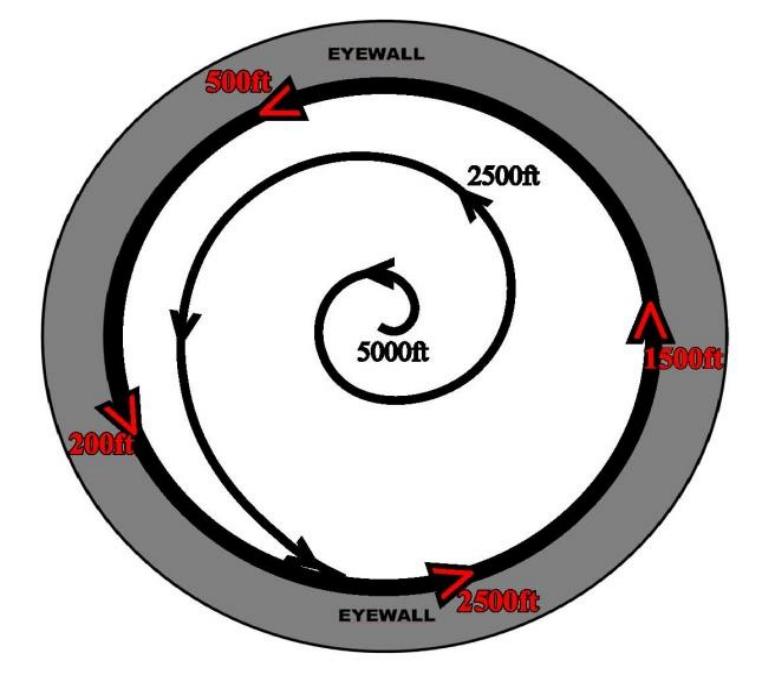
Hurricane Eye CONOP #2

Launch at 10,000 ft. Spiral descent to 2500 ft.
Then, enter the eyewall and step descend to 200 ft.
After 200 ft, it will maintain that altitude until the aircraft is destroyed from the environment or power is lost.

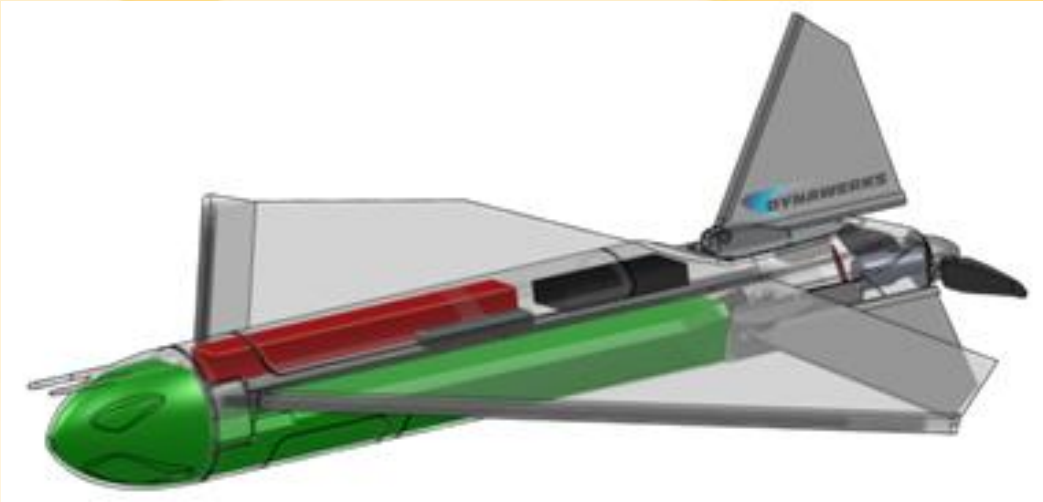
P3 – Flight Path

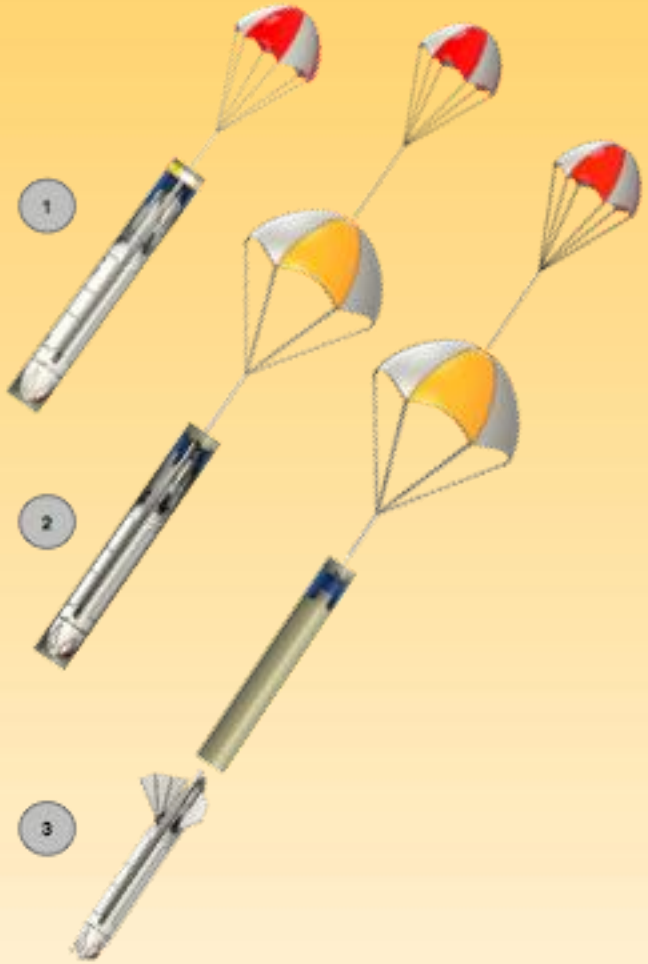


Gale UAS – Flight Path

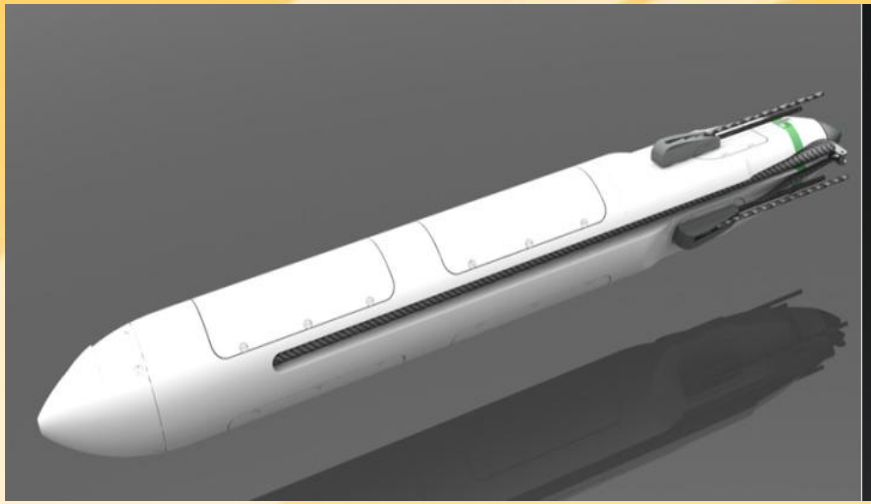


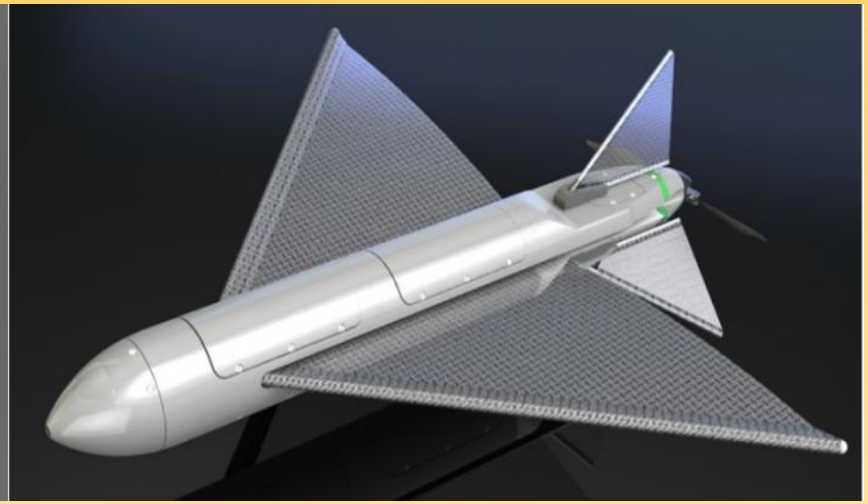
Deployment





Airframe and Aircraft Design



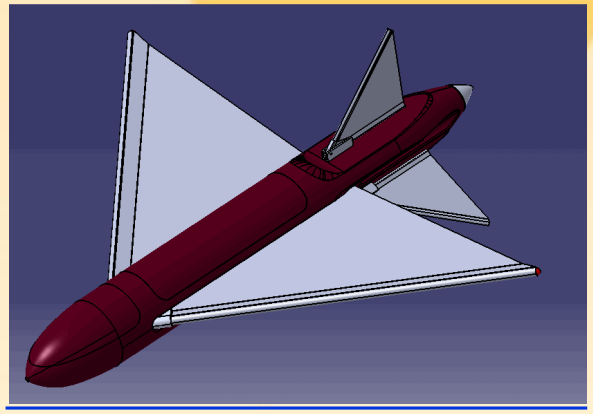


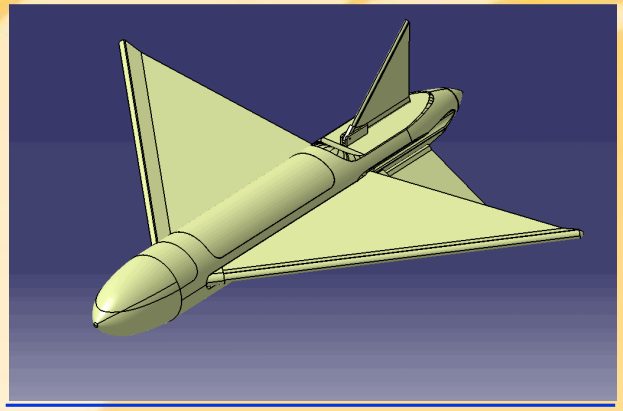
Attribute	Specification
Wingspan	42 in.
Length	35 in.
Fuselage Diameter (outer)	4.6 in.
Mission Weight	8.0 lbm
Cruise Speed	42 kts
Dash Speed	110 kts
Stall Speed	22 kts
Mission Endurance	60 minutes

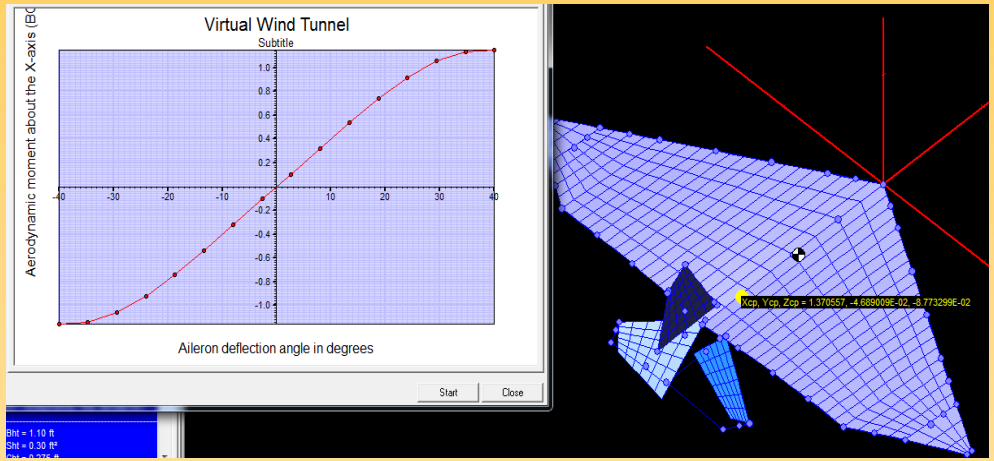
Design evolution from:

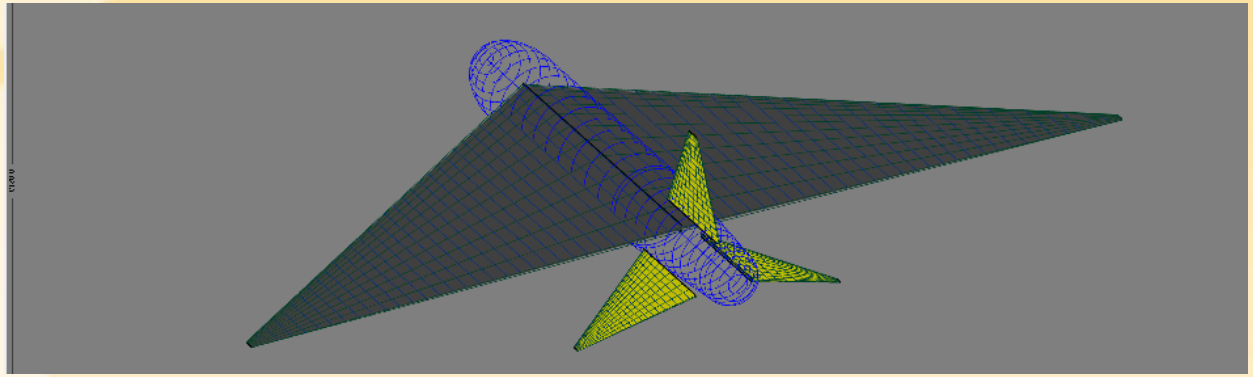
- CAD and CFD analysis
- Wind tunnel analysis
- Avionics and Payload integration
- Ground tests
- Ground launch
- “Dress Rehearsal” P3 deployment

CFD and CAD Analysis

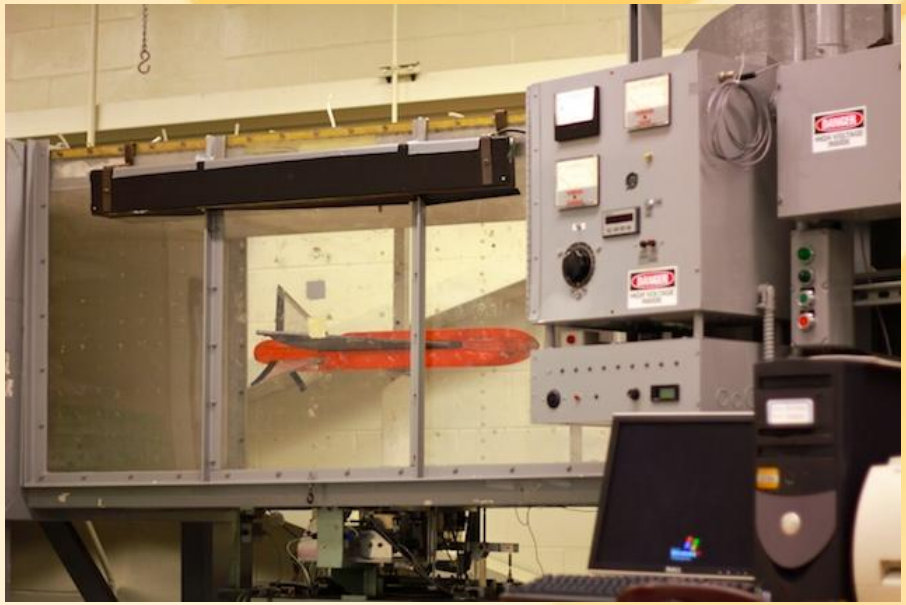








Wind Tunnel Analysis



Repair of Stress Factures Issues

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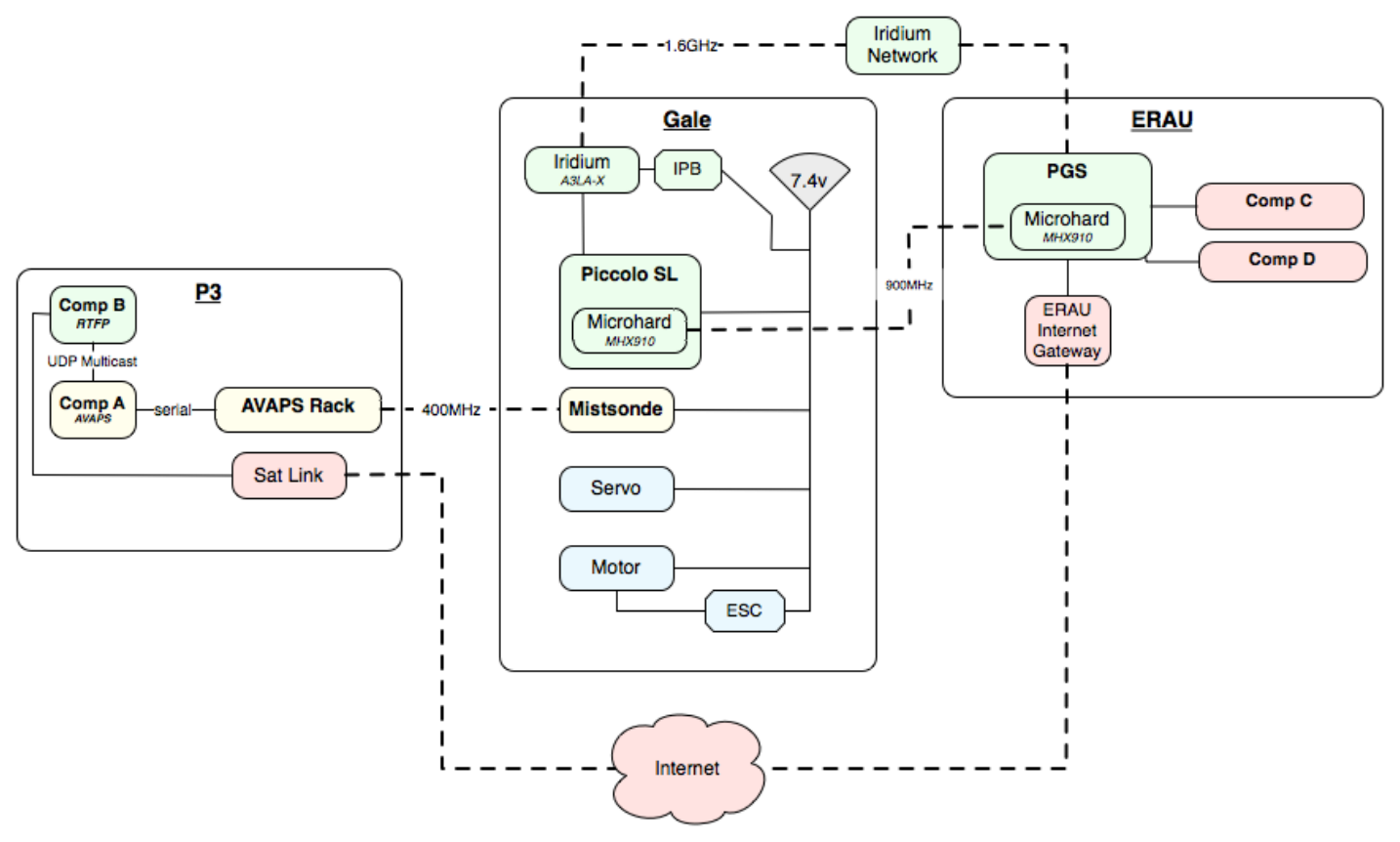
Problems

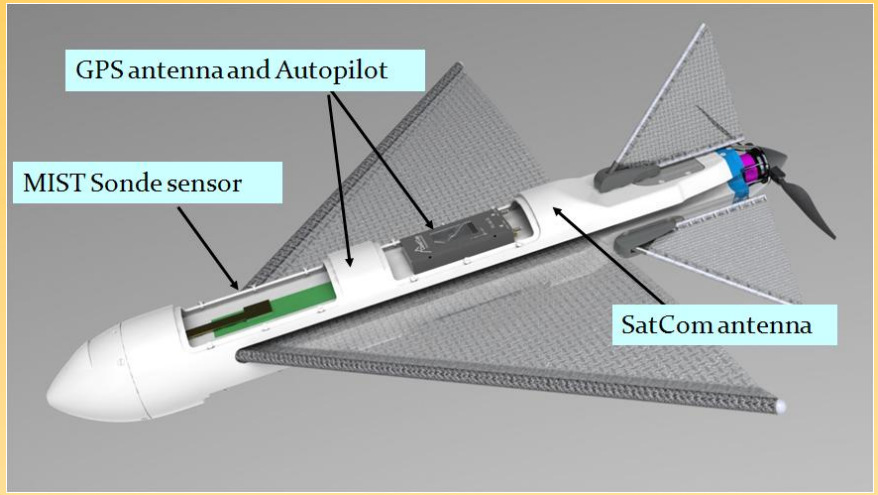
- Repeated insertion and removal of Gale from sabot tube resulted in some fatigue damage
- When collapsed, tensile and shear stresses applied to screws and mounting plates

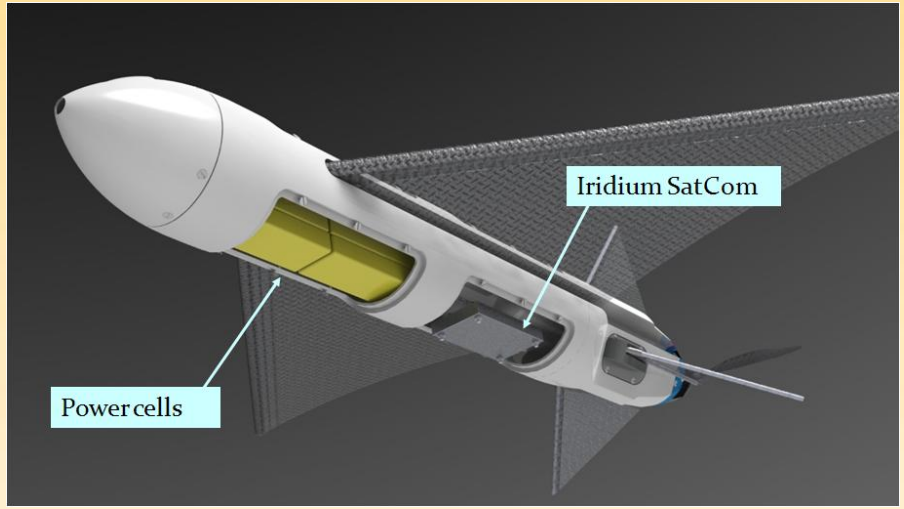
Solutions

- Redesigned mounting plates out of aluminum
- Latch points added to body to ensure screws secure fit of mounting screws
- Note: aircraft weight increased as a result
- Resulted in cancelation of 2011 clear air flight test and 1-year to accommodate re-design and new flight window

Avionics and Communications







Piccolo SL Autopilot

EMBRY
Aeronauti

Software Options

Standard Feature Set + Peripherals: Adds new support for pan-tilt servos, improved GPS/INS performance, and more flexibility in configuring payload ports.

Laser Altimeter Autoland: Provides accurate altitude information allowing the vehicle to perform a soft flared landing (Laser altimeter hardware sold separately).

DGPS Autoloand: Extends the autoland performance by using 2 cm accuracy DGPS. Supports autonomous taxi, rolling take-off, stationary and moving net recovery. Uses NovAtel DGPS equipment.

DGPS and Moving Net Recovery: Adds support of moving net recovery needed for shipboard and other moving capture applications. Uses NovAtel DGPS equipment and associated antennas.

Helicopter Operations (VTOL): Includes take-off and landing, precision hover, and automated path following along with autopilot assisted manual steering modes.

Radio Options

900 MHz unlicensed ISM

900 MHz Australian band

2.4 GHz unlicensed ISM

310-390 MHz discrete

1350-1390 MHz discrete

1670-1700 MHz discrete

Specifications

EMI shielded aluminum enclosure

RS232 Payload interface: 3

Fourteen (14) configurable GPIO lines. Four GPIO lines can be configured as analog inputs, 0-5V input, 10 bit conversion

CAN: Simulation / General interface

Flight termination: Deadman output

Integrated RF data link options: 900 MHz unlicensed ISM. 900 MHz Australian band. 2.4 GHz unlicensed ISM. 310-390 MHz discrete. 1350-1390 MHz discrete. 1670-1700 MHz discrete

GPS: 4 Hz uBlox module GPS receiver, 5 volt

Pressure Sensors: Ported static. 15-115 KPa-ported pitot. 6 KPa differential. 192 kts max indicated airspeed

Waypoint navigation: 1000 waypoints saved in autopilot

Inertial Sensors: 3 axis gyroscopes, 300°/sec. 3 axis acceleration, 6g

Supported peripherals: Transponders, secondary comms radios, Iridium SatComm, TASE gimbals, servo PTZ gimbals, magnetometers, laser altimeters, payload passthrough, RTK GPS

Vin: 4.5 – 28 volts

Power: 4 W (typical including 900 MHz radio)

Size: 131 x 57x 19 mm (5.1 x 2.24 x 0.75 inches)

Weight: 110 grams (3.9 oz) with 900 MHz radio

Operating temperature: -40°C to +80°C (calibrated range with case)

Power Requirements

Worst Case

System	Power (W)
Autopilot	3
Iridium	30
MIST Sonde	5
Servos	44
Motor	200
Misc	15
Total	297

Typical Case

System	Power (W)
Autopilot	3
Iridium	6
MIST Sonde	5
Servos	4
Motor	140
Misc	2
Total	160

Batteries

Type	Specific Energy (Wh/kg)	Advantages	Disadvantages
LiPo	170	Lighter compared to $LiFePO_4$, highest nominal voltage	Expensive and easy to damage
$LiFePO_4$	120	Fast charge cycle, excellent durability, and the safest	Low energy density and the heaviest
LiS	>200	Highest energy density and lighter than all other available batteries	No commercial availability and requires more cells
$LiSOCl_2$	>400	One time use and highest energy density of all lithium battery types	High cost, many safety concerns, and mainly used in military applications

Note: LiPo: Lithium Polymer
$LiFePO_4$: Lithium Iron Phosphate
LiS: Lithium Sulphate
$LiSOCl_2$: Lithium Thionyl Chloride

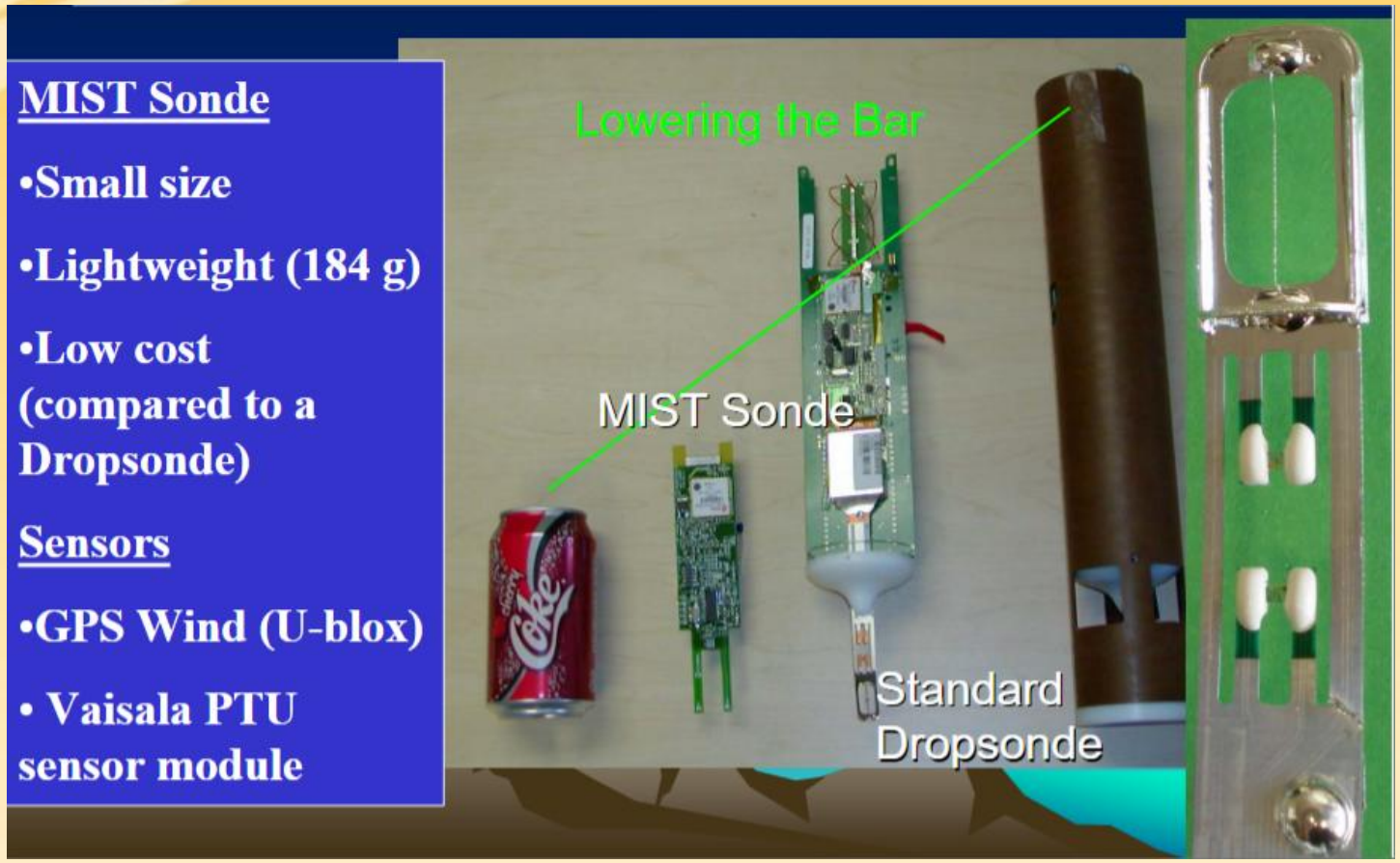
GPS Dropsonde Payload

MIST Sonde

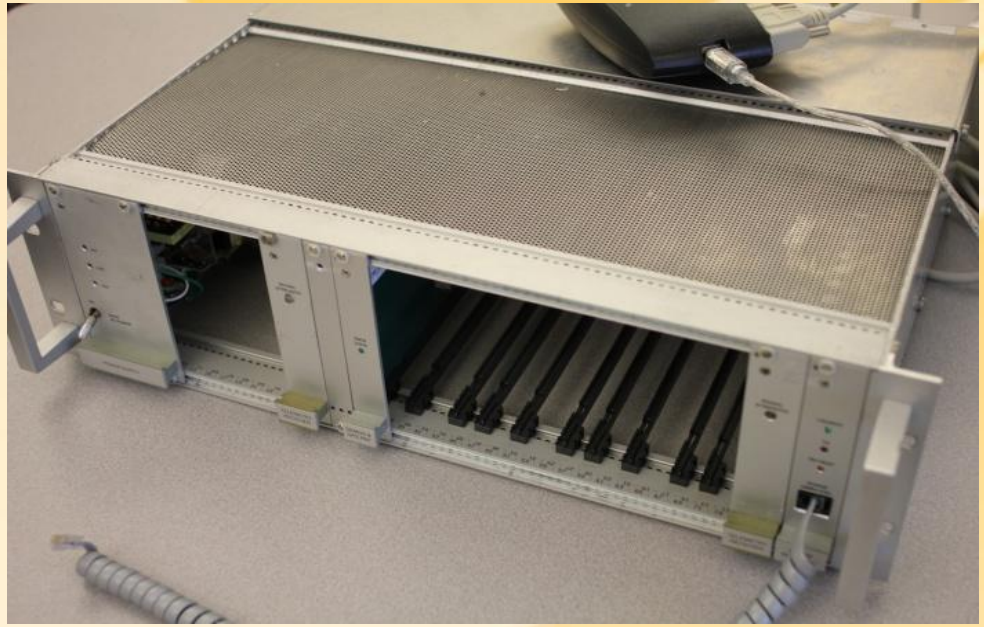
- Small size
- Lightweight (184 g)
- Low cost (compared to a Dropsonde)

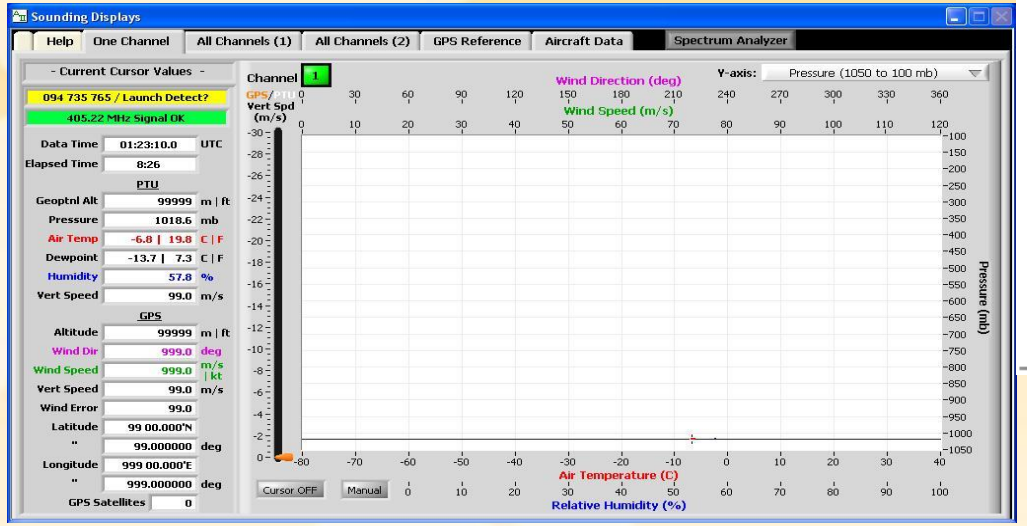
Sensors

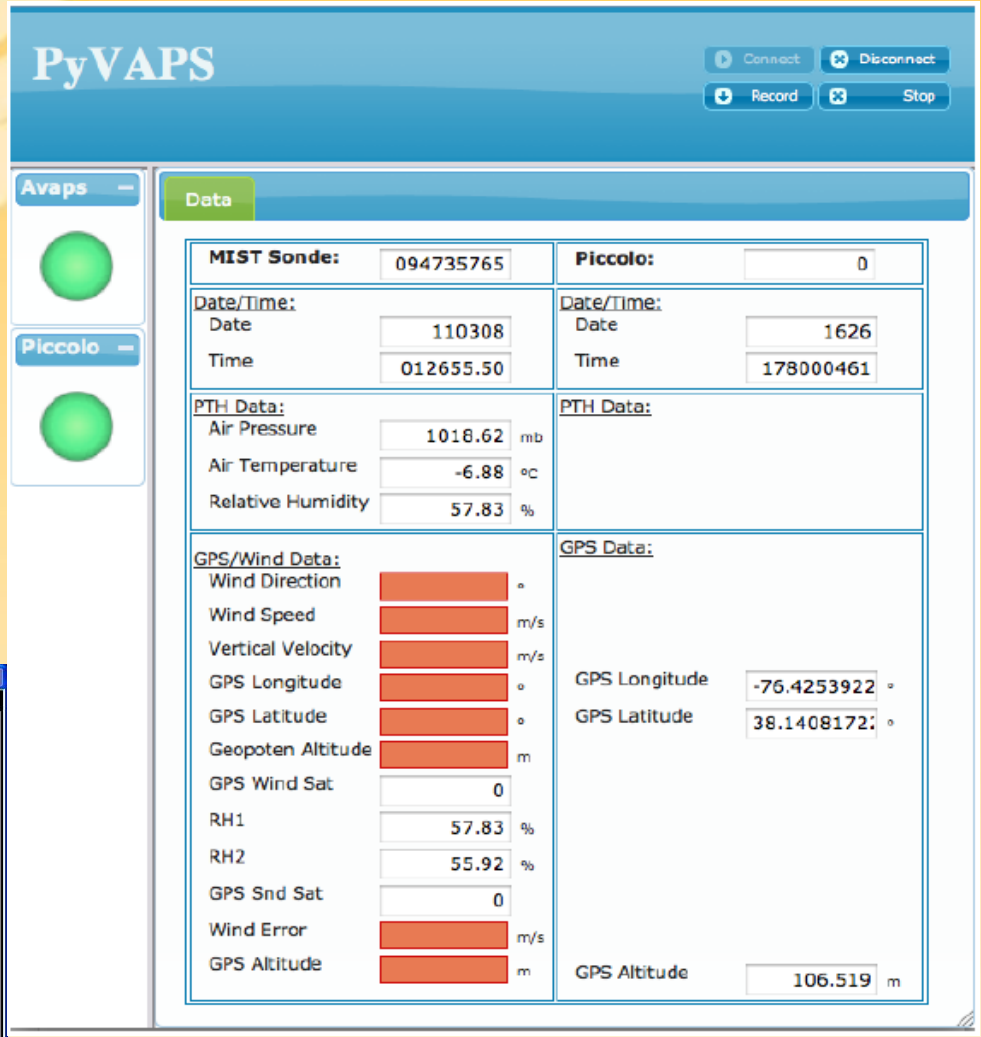
- GPS Wind (U-blox)
- Vaisala PTU sensor module



Data Collection and Processing





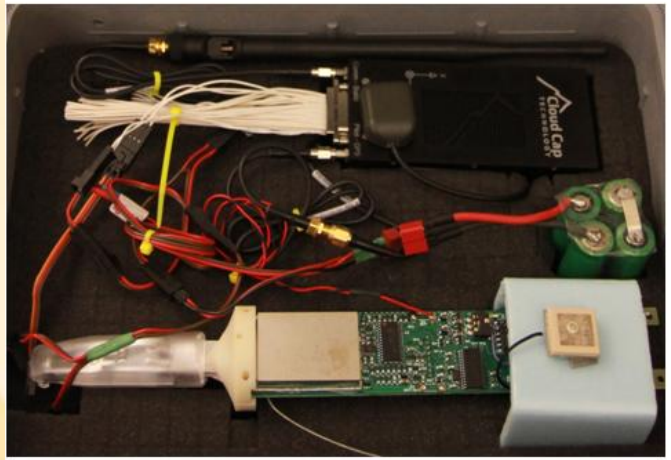


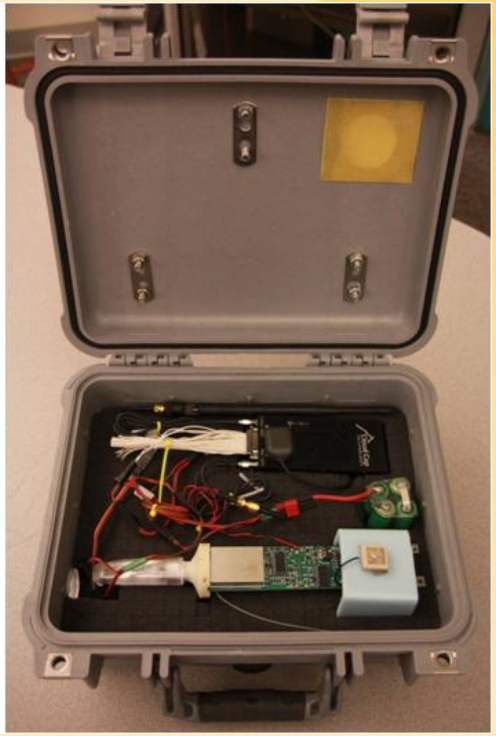
Sensor fusion required:

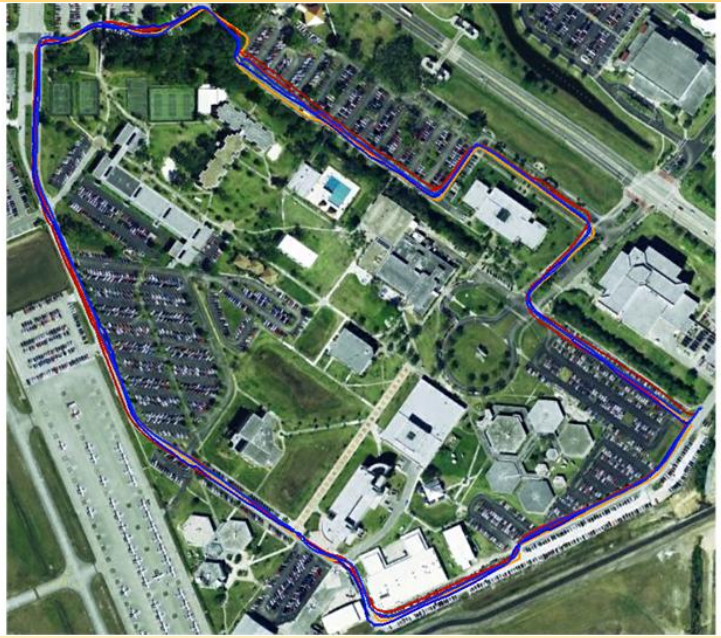
MIST Sonde (via AVAPS)
- Temperature
- Pressure
- Humidity

Piccolo SL (via C2 RF LOS/BLOS data link)
- GPS
- Wind estimation data

Ground Tests of Payload





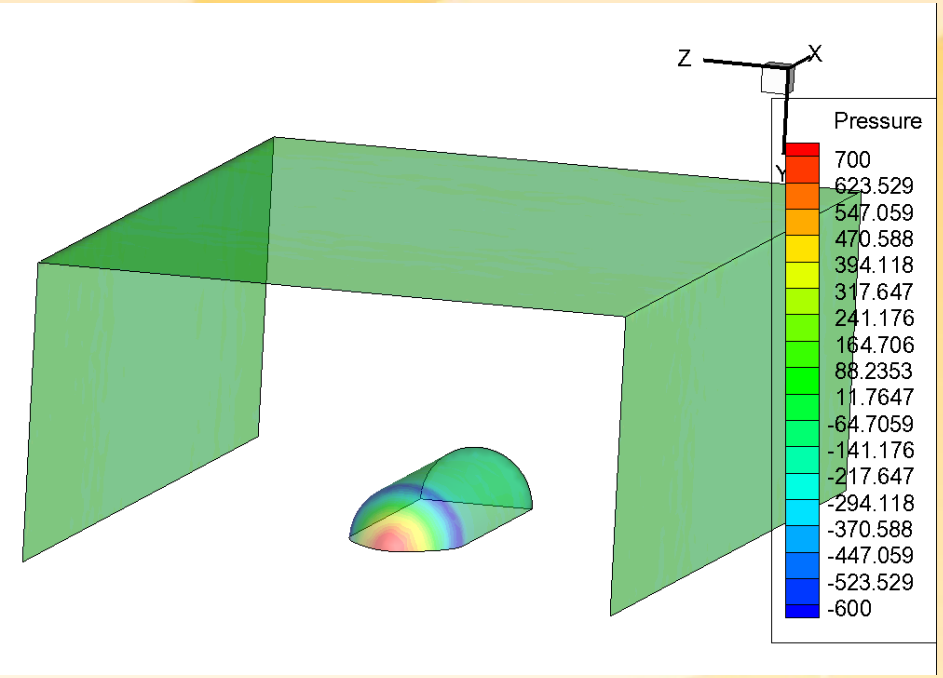


Car Path

Sonde Paths

Piccolo Paths

Integration Prototyping



Deployment

- Sabot Tube
  - Parachute held on by a plastic flap
  - When entering wind stream, the flap is pulled away deploying the parachute.
- Process
  - Deployment
  - Sabot Tube Pulled Away
  - Falls like a dart (ballistic)
  - Wings deploy and lock into place
  - Automation takes over flight controls
    - Remote pilot can also control during clear air flights
- All of this done using timers and configured for 10,000 ft deployment.

Ground Launch Test

Build – Test - Fly

Ground testing provided an ability to prove flight concept:

- Tune autopilot
- Identify issues with current model
- Initially problems encountered with torque roll upon liftoff

Vehicle launch proved problematic

- Issues of damage during control system tuning

Transitioned to a tensioned line launch

Vehicle inherently difficult to launch as designed for operation at higher altitudes and gradual descent. Not designed for climb.

Note: unable to secure non-P3 option for airborne deployment test

Torque Roll and Yaw Stability

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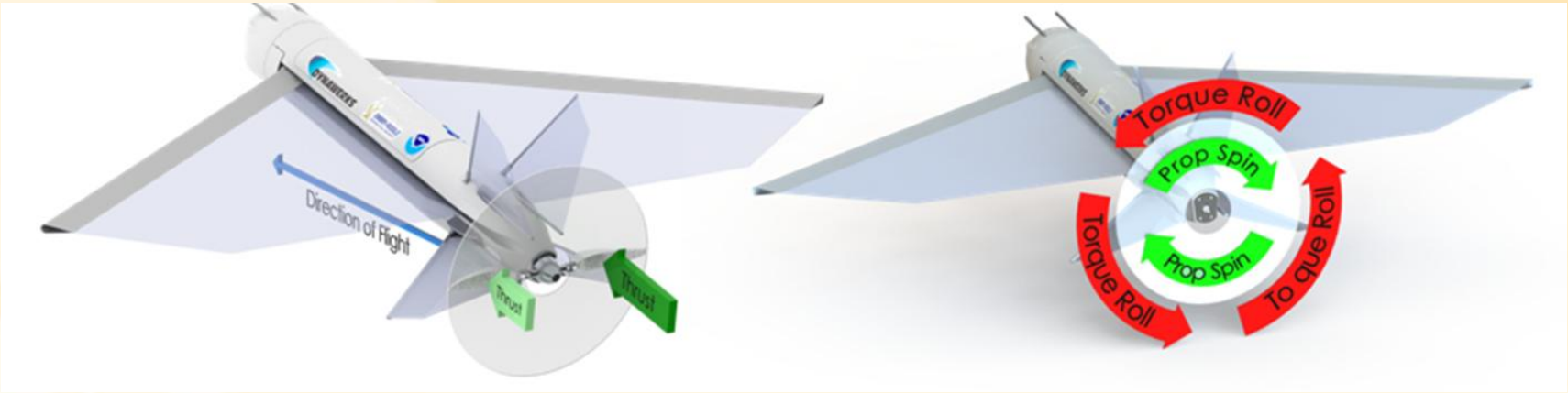
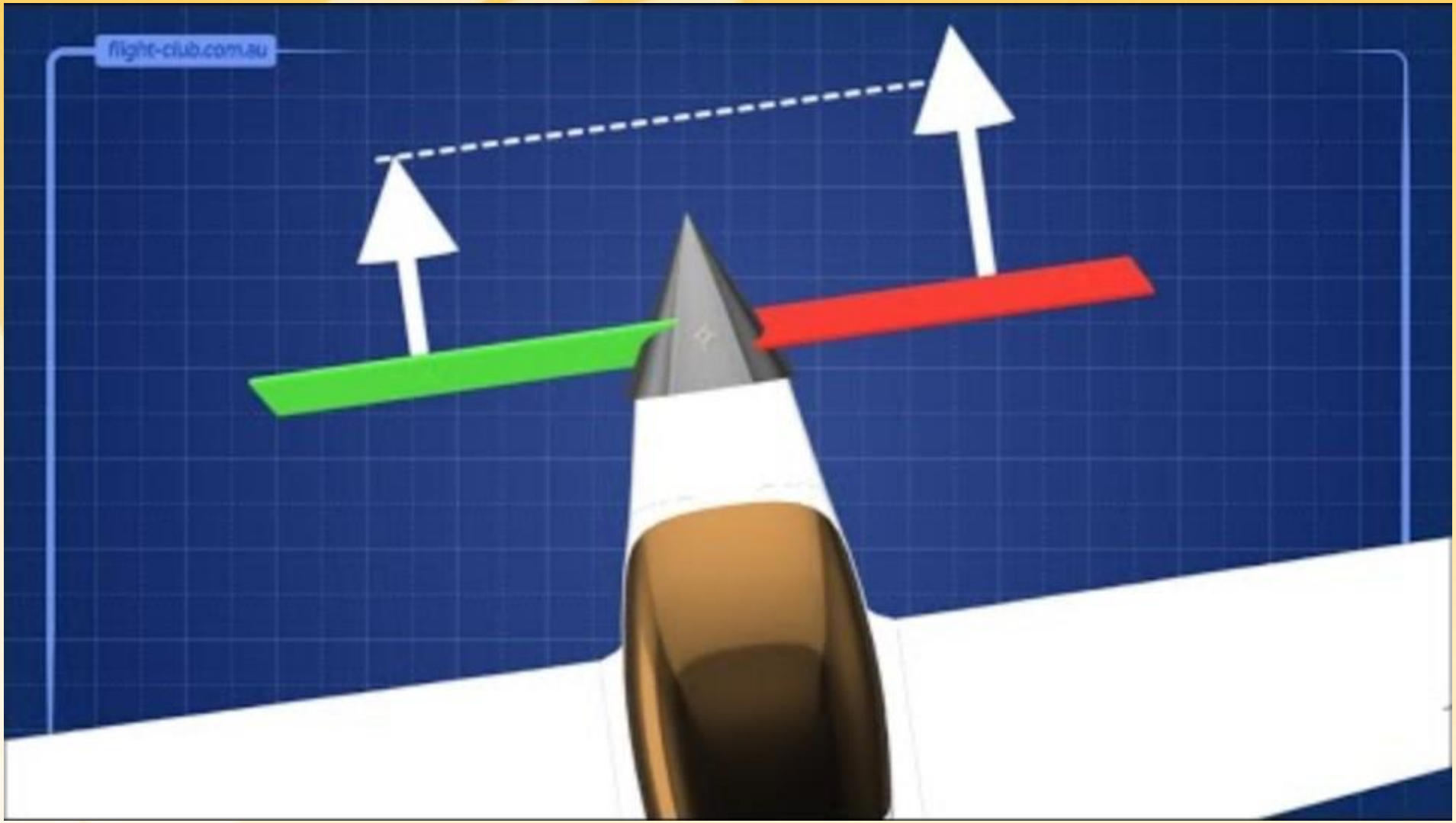
Problems

- Analysis determined two stability control issues as a result of added aircraft weight
  - Increased torque roll
  - Increased P-factor
    - Due to high angle of attack

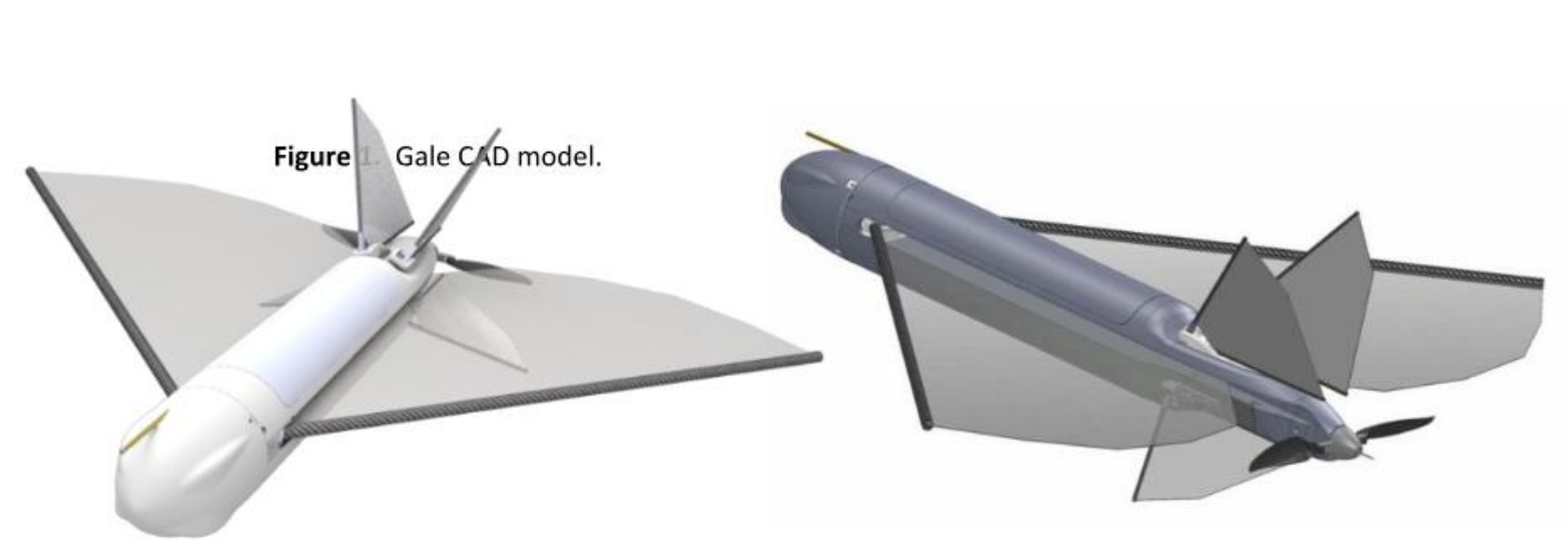
Solutions

- CFD, wind tunnel, and direct analysis
- Constructed model of aircraft for HIL testing for autopilot tuning
- Second vertical stabilizer added to improve lateral stability




flight-club.com.au

Airframe Updated



Gale Flies!

Clear Air Flight - Overview

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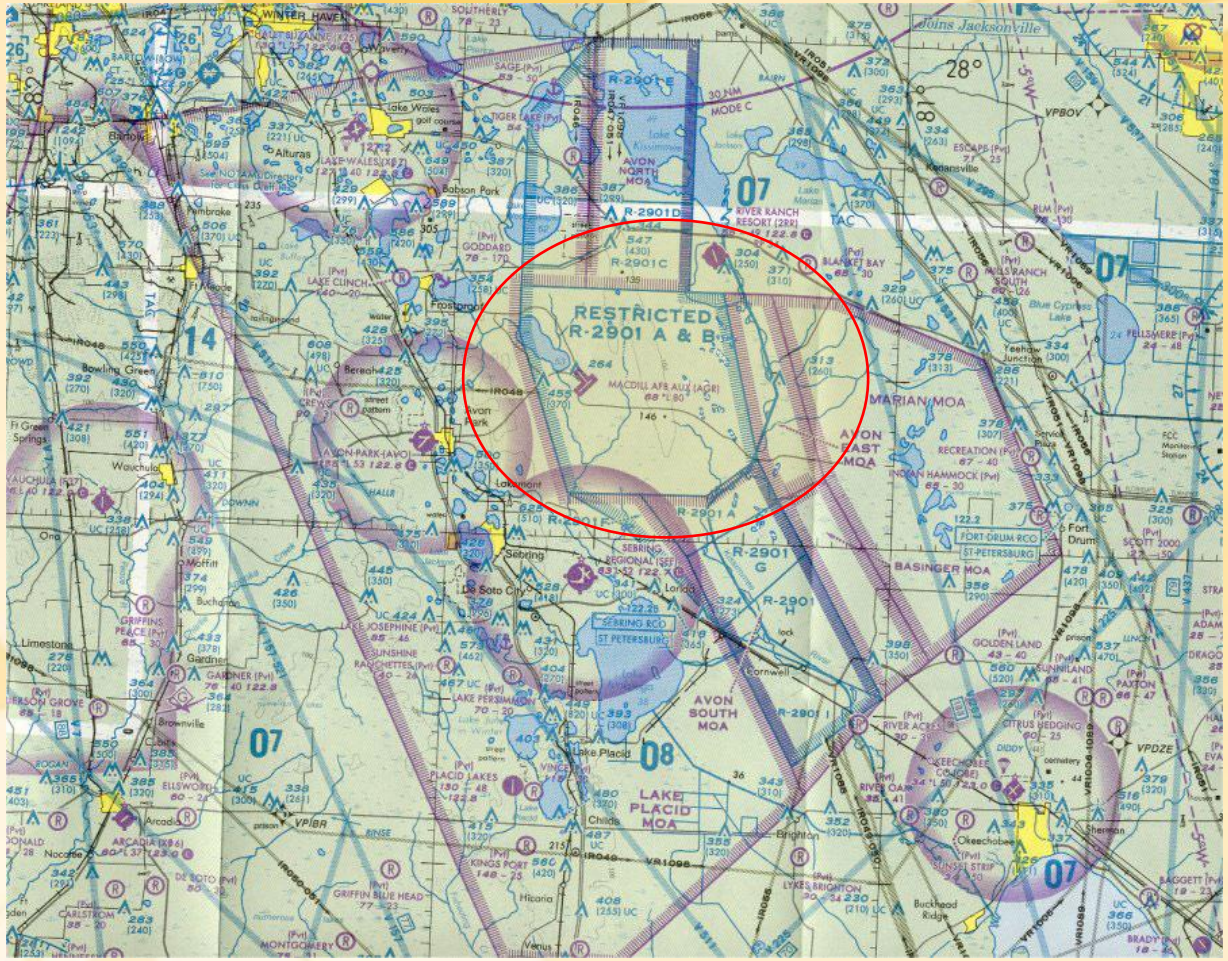
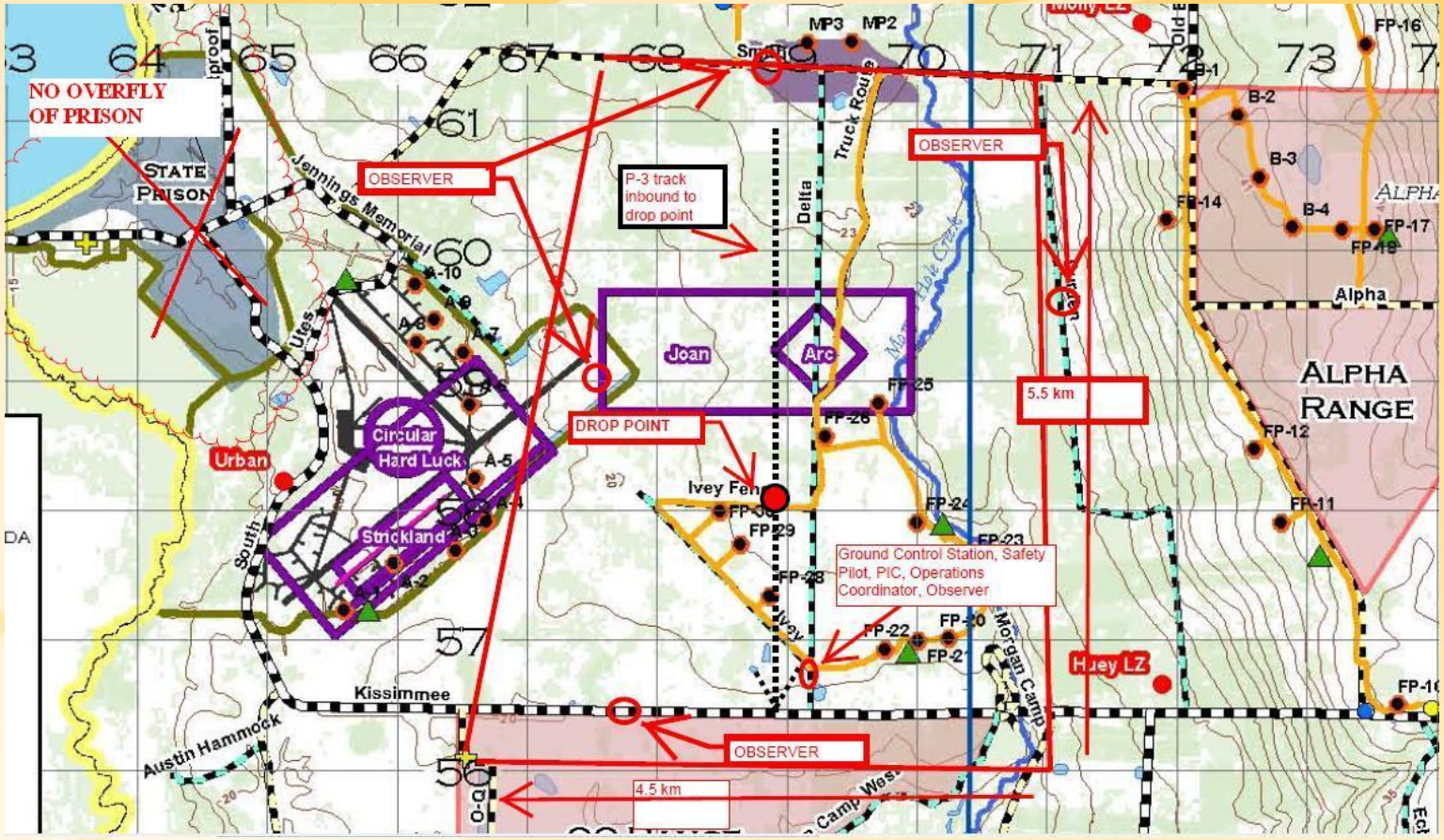
- Prior to operation within a tropical cyclone a clear air flight is required
- Goals
  - End-to-end pre-flight analysis onboard P-3 (RFI, AVAPS testing, fit check, etc.)
  - Full, but abbreviated, mission profile execution
  - Validation of science payload using GPS dropsondes
- Success criteria
  - Collected data meets the payload performance requirements
  - Aircraft completes entire mission
- Preparations
  - Site selection
    - AVON Park MOA
  - Final airworthiness review by NOAA engineers
    - Requirements inspection
    - Design inspection
    - Vehicle inspection
  - Mission CONOP Development
    - Waypoints
    - Geofence
    - Contingency management plan
    - Checklists
  - Water survival training for ERAU personnel on P-3
  - Final mission brief with flight crew

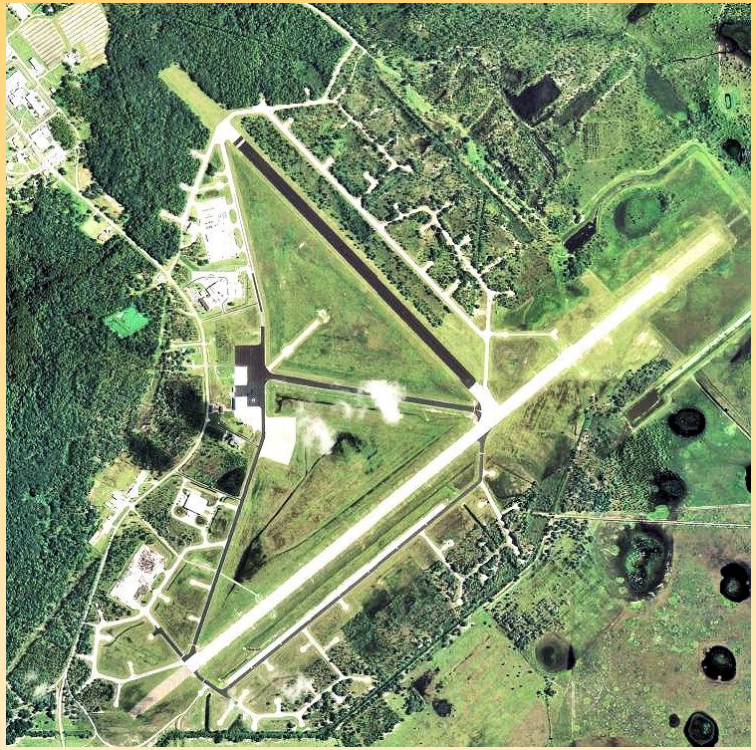
NOAA Sky Hopper WD-P3

Avon Park MOA

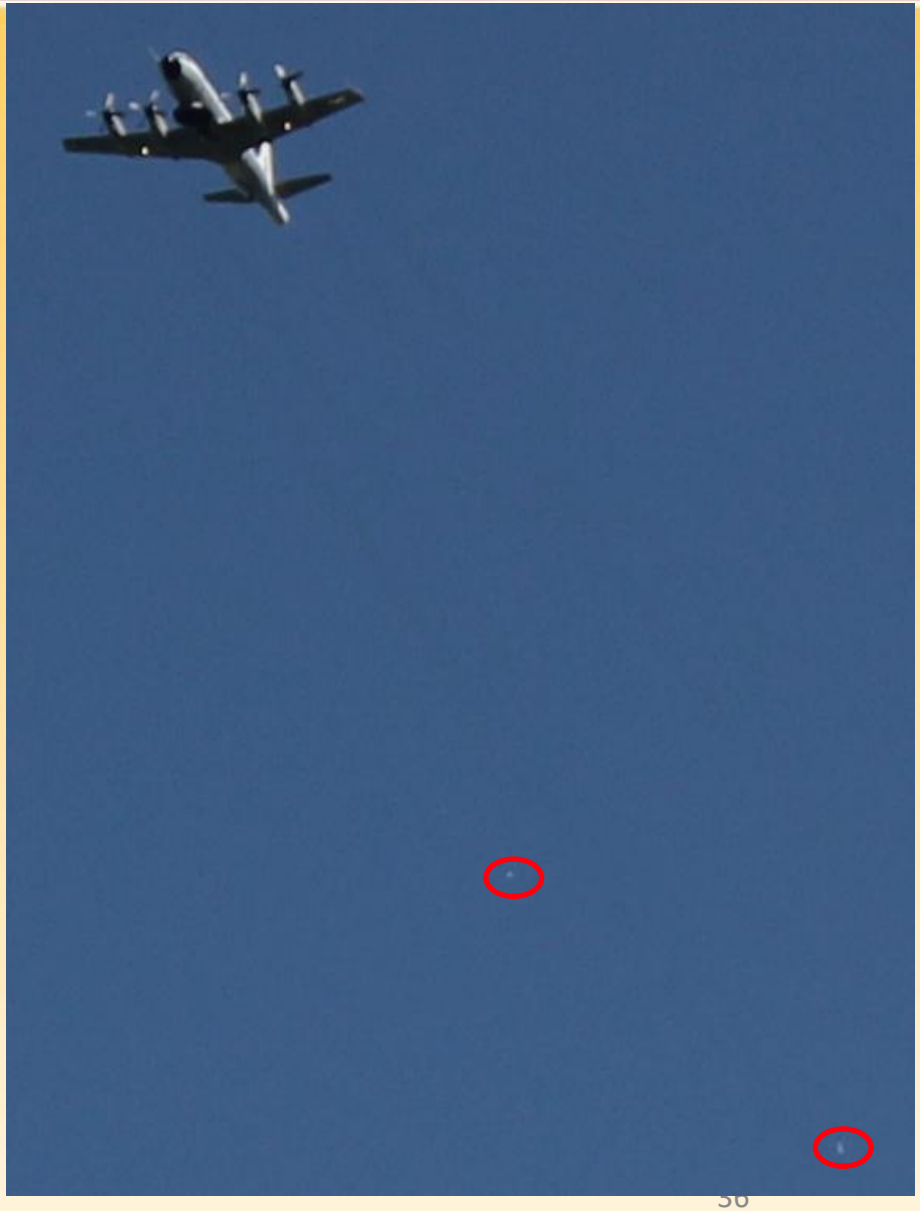


Detailed View of Airspace Configuration





Clear Air Flight



Results

- Deployment successful
- Only 90% of wing deployment (technical fault)
- UAS climb-out of dive started
- Ground impact not observed
  - Unable to recovery from ballistic trajectory and enter flight regime
  - Belly strike
  - Impact with palmetto on ground ruptured battery pack resulting in fire
  - Total hull loss ☹

Post-Flight System Upgrades

- New release mechanism
  - Once sabot tube reaches terminal velocity under drogue chute, released.
  - Gale tethered in tube by wire. Wire heated to break away and release for flight
- Lower cost autopilot
  - Pixhawk Autopilot
  - Significantly lower cost compared with Piccolo SL
- Airframe redesign
  - Lighter airframe
  - 3D printing using biodegradable polymers
- Waiting for future flight opportunities

Other Hurricane UAS - Coyote

Aspect	Details
(a) System Developer	Raytheon (acquired from BAE Systems Inc)
(b) Sponsor	NOAA (National Oceanic and Atmospheric Administration)
(c) Storm Flown	* Hurricane Edouard (2014), * Maria (2017) and * Michael (2018)
(d) Date Flown	September 2014 (Edouard), 2017 (Maria), 2018 (Michael)
(e) Summary of Results	• The Coyote was air-launched from a NOAA P-3 Orion research plane. It collected wind, humidity, and pressure readings in the intense section of the hurricane

Coyote

Other Hurricane UAS: Altuis-600

Aspect	Details
(a) System Developer	Black Swift Technologies
(b) Sponsor	NOAA (National Oceanic and Atmospheric Administration)
(c) Storm Flown	* Hurricane Helene * Tropical Storm Tammy * Hurricane Ernesto * Hurricane Ian
(d) Date Flown	* Hurricane Helene: August 2024 * Tropical Storm Tammy: 2023 * Hurricane Ernesto: 2024 * Hurricane Ian: 2022
(e) Summary of Results	• Hurricane Ian: The Altius-600 drone was flown for the first time in Hurricane Ian in 2022, and a team from NOAA's Aircraft Operations Center and its Acquisition and Grants Office won a Department of Commerce Gold Medal for that work. • Tropical Storm Tammy: Used to measure parts of the storm too dangerous for humans to go. • Hurricane Helene: Successfully launched from a NOAA P-3 aircraft to sample the eye, eyewall, and storm's lowest layers, breaking records in flight time, communications range, and maximum recorded wind speed.

Altius-600